Deploy de VM en Azure con Terraform (Costo Minimo)

1. main.tf (Archivo Terraform)

```
provider "azurerm" {

  features {}

}


resource "azurerm_resource_group" "rg" {

  name     = "rg-terra-vm"

  location = "East US"

}


resource "azurerm_virtual_network" "vnet" {

  name                = "vnet-terra"

  address_space       = ["10.0.0.0/16"]

  location            = azurerm_resource_group.rg.location

  resource_group_name = azurerm_resource_group.rg.name

}


resource "azurerm_subnet" "subnet" {

  name                 = "subnet-terra"

  resource_group_name  = azurerm_resource_group.rg.name

  virtual_network_name = azurerm_virtual_network.vnet.name

  address_prefixes     = ["10.0.1.0/24"]

}
```

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```
resource "azurerm_public_ip" "public_ip" {

  name                = "pip-terra"

  location            = azurerm_resource_group.rg.location

  resource_group_name = azurerm_resource_group.rg.name

  allocation_method   = "Dynamic"

}


resource "azurerm_network_interface" "nic" {

  name                = "nic-terra"

  location            = azurerm_resource_group.rg.location

  resource_group_name = azurerm_resource_group.rg.name


  ip_configuration {

    name                          = "internal"

    subnet_id                     = azurerm_subnet.subnet.id

    private_ip_address_allocation = "Dynamic"

    public_ip_address_id          = azurerm_public_ip.public_ip.id

  }

}


resource "azurerm_linux_virtual_machine" "vm" {

  name                = "vm-terra"

  resource_group_name = azurerm_resource_group.rg.name

  location            = azurerm_resource_group.rg.location

  size                = "Standard_B1s"
```

```
  admin_username = "azureuser"

  network_interface_ids = [

    azurerm_network_interface.nic.id,

  ]



  admin_ssh_key {

    username   = "azureuser"

    public_key = file("~/.ssh/id_rsa.pub")

  }



  os_disk {

    caching              = "ReadWrite"

    storage_account_type = "Standard_LRS"

  }



  source_image_reference {

    publisher = "Canonical"

    offer     = "UbuntuServer"

    sku       = "18.04-LTS"

    version   = "latest"

  }

}



output "public_ip_address" {

  value = azurerm_public_ip.public_ip.ip_address

}
```

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2. Procedimiento Paso a Paso

a. Instalar Terraform y Azure CLI

```
# Terraform
sudo apt-get update && sudo apt-get install -y gnupg software-properties-common
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /etc/apt/trusted.gpg.d/hashicorp-archive-keyring.gpg
echo "deb [signed-by=/etc/apt/trusted.gpg.d/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install terraform

# Azure CLI
curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
```

b. Login en Azure y Configurar Credenciales

```
az login
az account list --output table
az account set --subscription "ID_O_NOMBRE_SUSCRIPCION"
```

c. Crear Service Principal para Terraform az ad sp create-for-rbac --name "terraform-sp" --role="Contributor" --scopes="/subscriptions/<SUBSCRIPTION_ID>"

d. Exportar las variables de entorno

export ARM_CLIENT_ID="appId"

export ARM_CLIENT_SECRET="password"

export ARM_SUBSCRIPTION_ID="subscriptionId"

export ARM_TENANT_ID="tenantId"

e. Inicializar y Aplicar con Terraform

terraform init

terraform plan

terraform apply

f. Conectar a la VM

ssh azureuser@<PUBLIC_IP>

3. Requisitos Previos

- Tener SSH Key (~/.ssh/id_rsa.pub)

- Azure CLI instalado

- Terraform instalado

Deploy de VM en Azure con Terraform (Costo Minimo)

- Tener Service Principal configurado

- Exportar las variables de entorno antes de usar Terraform